PRIME-PAIR KERNELS, MANGOLDT RESONANCES, AND A FINITE-HEIGHT ENERGY BRIDGE

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1. Shells and normalized kernels

Write $\mathbb{N}_0 = \{0, 1, 2, \ldots\}$. Let

$$\mathcal{S}_R \subset \mathbb{N}_0^2 \qquad (R > 0) \tag{1.1}$$

be finite nested sets satisfying

$$\mathcal{S}_R \subset \mathcal{S}_{R'} \quad (R < R'), \qquad \bigcup_{R>0} \mathcal{S}_R = \mathbb{N}_0^2. \tag{1.2}$$

The parameter R indexes the arithmetic shell; the letter T will be reserved for the height of zeta zeros.

Fix distinct primes p, q. Define

$$W_R^{(p,q)} = \sum_{(a,b)\in\mathcal{S}_R} p^{-a} q^{-b}, \tag{1.3}$$

$$S_R^{(p,q)} = \sum_{(a,b)\in\mathcal{S}_R} p^{-2a} q^{-2b}, \tag{1.4}$$

$$w_{a,b;R}^{(p,q)} = \frac{p^{-a} q^{-b}}{W_R^{(p,q)}} \qquad ((a,b) \in \mathcal{S}_R). \tag{1.5}$$

Then the weights are positive and have sum one.

Proposition 1.1 (Coefficient energy)**.** *Set*

$$Q_R^{(p,q)} = \sum_{(a,b)\in\mathcal{S}_R} \left(w_{a,b;R}^{(p,q)}\right)^2 = \frac{S_R^{(p,q)}}{\left(W_R^{(p,q)}\right)^2}. \tag{1.6}$$

Then

$$Q_R^{(p,q)} \longrightarrow Q_\infty^{(p,q)} = \frac{p-1}{p+1}\frac{q-1}{q+1} \in (0,1). \tag{1.7}$$

Proof. Monotone convergence of the two geometric series gives

$$W_R^{(p,q)} \longrightarrow \frac{1}{(1-p^{-1})(1-q^{-1})}, \qquad S_R^{(p,q)} \longrightarrow \frac{1}{(1-p^{-2})(1-q^{-2})}. \tag{1.8}$$

Substitution in (1.6) gives (1.7). □

For $\vartheta = (\vartheta_p, \vartheta_q) \in \mathbb{T}^2 = (\mathbb{R}/2\pi\mathbb{Z})^2$, put

$$u_{a,b}^{(p,q)} = a \log p + b \log q \tag{1.9}$$

and define

$$K_{R,\vartheta}^{(p,q)}(z) = \sum_{(a,b)\in\mathcal{S}_R} w_{a,b;R}^{(p,q)} e^{-i(a\vartheta_p + b\vartheta_q)} e^{izu_{a,b}^{(p,q)}}. \tag{1.10}$$

2. Limit kernel and pointwise resonance

Define

$$w_{a,b;\infty}^{(p,q)} = (1-p^{-1})(1-q^{-1})p^{-a}q^{-b}. \tag{2.1}$$

Proposition 2.1 (Uniform convergence on the real axis)**.** *For every fixed $\vartheta \in \mathbb{T}^2$,*

$$K_{R,\vartheta}^{(p,q)}(\tau) \longrightarrow K_{\infty,\vartheta}^{(p,q)}(\tau) \tag{2.2}$$

uniformly for $\tau \in \mathbb{R}$, where

$$K_{\infty,\vartheta}^{(p,q)}(z) = \sum_{a,b\geq 0} w_{a,b;\infty}^{(p,q)} e^{-i(a\vartheta_p + b\vartheta_q)} e^{izu_{a,b}^{(p,q)}} \tag{2.3}$$

$$= \frac{(1-p^{-1})(1-q^{-1})}{\left(1-p^{-1}e^{i(z\log p - \vartheta_p)}\right)\left(1-q^{-1}e^{i(z\log q - \vartheta_q)}\right)}. \tag{2.4}$$

The series is locally uniformly convergent in the half-plane $\operatorname{Im} z > -1$.

Proof. Extend the finite weight vector by zero outside $\mathcal{S}_R$. The exhaustion and the convergence of $W_R^{(p,q)}$ imply

$$\sum_{a,b\geq 0} \left| w_{a,b;R}^{(p,q)} - w_{a,b;\infty}^{(p,q)} \right| \longrightarrow 0. \tag{2.5}$$

This proves uniform convergence on the real axis. The geometric series give (2.4); the same argument on compact subsets of $\operatorname{Im} z > -1$ proves local uniform convergence there. □

Proposition 2.2 (Pointwise resonance)**.** *For real τ,*

$$\left| K_{\infty,\vartheta}^{(p,q)}(\tau) \right| \leq 1. \tag{2.6}$$

Moreover,

$$\sup_{\tau\in\mathbb{R}} \left| K_{\infty,\vartheta}^{(p,q)}(\tau) \right| = 1. \tag{2.7}$$

For $\vartheta = (0,0)$ equality occurs at $\tau = 0$.

Proof. For $0 < r < 1$ and real t,

$$|1 - re^{it}| \geq 1 - r. \tag{2.8}$$

Applying this to both factors in (2.4) proves (2.6). Unique factorization gives

$$m \log p + n \log q = 0, quad(m,n) \in \mathbb{Z}^2 \quad \Longrightarrow \quad m = n = 0. \tag{2.9}$$

Consequently the line $\tau(\log p, \log q)$ is dense modulo 2π in $\mathbb{T}^2$. It approaches the prescribed phase ϑ arbitrarily closely, proving (2.7). □

3. Equidistribution of zeta-zero phases

Let $\rho = \beta + i\gamma$ run through the nontrivial zeros of ζ, with $0 < \gamma \le T$, counted with multiplicity, and let $N(T)$ be their number.

Theorem 3.1 (Joint zero-phase equidistribution)**.** *For distinct primes p, q, the vectors*

$$(\gamma \log p, \gamma \log q) \pmod{2\pi} \tag{3.1}$$

are uniformly distributed in $\mathbb{T}^2$ as $T \to \infty$.

Proof. Hlawka's theorem states that, for every fixed nonzero real number α, the sequence $\alpha\gamma$ is uniformly distributed modulo one; see [1]. For every $(m, n) \in \mathbb{Z}^2 \setminus \{(0, 0)\}$, equation (2.9) gives

$$\frac{m \log p + n \log q}{2\pi} \neq 0. \tag{3.2}$$

Hlawka's theorem applied to every nontrivial character, followed by the multidimensional Weyl criterion, proves the assertion. □

4. Exact mean-square contraction

Theorem 4.1 (Finite-shell contraction on the zero ordinates)**.** *For every fixed shell R and phase ϑ,*

$$\lim_{T\to\infty} \frac{1}{N(T)} \sum_{0<\gamma\le T} \left| K_{R,\vartheta}^{(p,q)}(\gamma) \right|^2 = Q_R^{(p,q)}. \tag{4.1}$$

Consequently,

$$\lim_{R\to\infty} \lim_{T\to\infty} \frac{1}{N(T)} \sum_{0<\gamma\le T} \left| K_{R,\vartheta}^{(p,q)}(\gamma) \right|^2 = Q_\infty^{(p,q)} < 1. \tag{4.2}$$

Proof. Expanding the square in (4.1) gives diagonal terms

$$\sum_{(a,b)\in\mathcal{S}_R} \left(w_{a,b;R}^{(p,q)} \right)^2 = Q_R^{(p,q)} \tag{4.3}$$

and off-diagonal characters of the vector (3.1). Theorem 3.1 makes every off-diagonal average vanish. Proposition 1.1 then gives (4.2). □

Corollary 4.2 (Limit-kernel contraction)**.** *For every fixed phase ϑ,*

$$\lim_{T\to\infty} \frac{1}{N(T)} \sum_{0<\gamma\le T} \left| K_{\infty,\vartheta}^{(p,q)}(\gamma) \right|^2 = Q_\infty^{(p,q)}. \tag{4.4}$$

Proof. The uniform convergence in Proposition 2.1 permits passage from the finite kernels to the limit kernel inside the normalized average. □

Define the torus kernel

$$\mathcal{K}_{p,q}(x, y) = \frac{(1 - p^{-1})(1 - q^{-1})}{(1 - p^{-1}e^{ix})(1 - q^{-1}e^{iy})}. \tag{4.5}$$

Theorem 4.3 (Iteration with one fixed pair)**.** *For every integer $r \ge 1$,*

$$E_r^{(p,q)} := \lim_{T\to\infty} \frac{1}{N(T)} \sum_{0<\gamma\le T} \left| K_{\infty,\vartheta}^{(p,q)}(\gamma) \right|^{2r} \tag{4.6}$$

$$= \frac{1}{(2\pi)^2} \int_0^{2\pi} \int_0^{2\pi} |\mathcal{K}_{p,q}(x, y)|^{2r} \, \mathrm{d}x \, \mathrm{d}y. \tag{4.7}$$

Moreover,

$$E_1^{(p,q)} = Q_\infty^{(p,q)}, \qquad E_r^{(p,q)} \longrightarrow 0 \quad (r \to \infty). \tag{4.8}$$

Proof. Equation (4.7) follows from Theorem 3.1. Proposition 2.2 gives $|\mathcal{K}_{p,q}| \le 1$, with equality only at $(0,0)$ modulo 2π. This equality set has Haar measure zero. Dominated convergence proves the second assertion in (4.8). □

Corollary 4.4 (The pair $(5,7)$)**.** *For $p=5$ and $q=7$,*

$$K_{\infty,0}^{(5,7)}(\tau) = \frac{24/35}{(1-5^{-1}e^{i\tau\log 5})\,(1-7^{-1}e^{i\tau\log 7})} \tag{4.9}$$

and

$$\boxed{\lim_{T\to\infty}\frac{1}{N(T)}\sum_{0<\gamma\le T}\left|K_{\infty,0}^{(5,7)}(\gamma)\right|^2 = \frac{1}{2}.} \tag{4.10}$$

The moments with exponent $2r$ tend to zero in the iterated order displayed in Theorem 4.3.

Remark 4.5 (One prime versus one dilation). A single dilation is unitary. A normalized average over the powers of one prime is a different operator: its limiting mean-square factor on the zero ordinates is $(p-1)/(p+1) < 1$. These two statements must not be identified.

5. Mangoldt resonances and the linear bridge

For a nontrivial zero write

$$\delta_\rho = \beta - \frac{1}{2}. \tag{5.1}$$

For fixed T with $N(T) > 0$, define the existing finite-height exponential difference

$$F_T(y) = \frac{1}{N(T)}\sum_{0<\gamma\le T}\left(e^{(\delta_\rho+i\gamma)y} - e^{i\gamma y}\right). \tag{5.2}$$

Let $\Lambda(x)$ equal $\log\ell$ when $x=\ell^k$ is an integral prime power and equal zero for all other real $x>1$.

Proposition 5.1 (Mangoldt response of a finite shell)**.** *Assume $(0,0)\in\mathcal{S}_R$. Write $n_{a,b}=p^aq^b$ and $w_{0,0;R}=w_{0,0;R}^{(p,q)}$. Then*

$$\begin{aligned}\frac{1}{T}\left(\sum_{0<\gamma\le T}K_{R,\vartheta}^{(p,q)}(\gamma-i\delta_\rho) - w_{0,0;R}N(T)\right)\\ \longrightarrow -\frac{1}{2\pi}\sum_{\substack{(a,b)\in\mathcal{S}_R\\(a,b)\neq(0,0)}} w_{a,b;R}^{(p,q)}e^{-i(a\vartheta_p+b\vartheta_q)}\frac{\Lambda(n_{a,b})}{\sqrt{n_{a,b}}},\end{aligned} \tag{5.3}$$

$$\begin{aligned}\frac{1}{T}\left(\sum_{0<\gamma\le T}K_{R,\vartheta}^{(p,q)}(\gamma) - w_{0,0;R}N(T)\right)\\ \longrightarrow -\frac{1}{2\pi}\sum_{\substack{(a,b)\in\mathcal{S}_R\\(a,b)\neq(0,0)}} w_{a,b;R}^{(p,q)}e^{-i(a\vartheta_p+b\vartheta_q)}\frac{\Lambda(n_{a,b})}{\sqrt{n_{a,b}}}.\end{aligned} \tag{5.4}$$

If $a,b\ge 1$, then $\Lambda(p^aq^b)=0$; hence the Mangoldt main term is supported on the two prime-power axes.

Proof. For every fixed $x>1$, Landau's formula gives [2]

$$\sum_{0<\gamma\le T}x^\rho = -\frac{T}{2\pi}\Lambda(x) + O_x(\log T). \tag{5.5}$$

Since

$$K_{R,\vartheta}^{(p,q)}(\gamma - i\delta_\rho) = \sum_{(a,b)\in\mathcal{S}_R} w_{a,b;R}^{(p,q)} e^{-i(a\vartheta_p + b\vartheta_q)} n_{a,b}^{\rho-1/2}, \tag{5.6}$$

finite summation of (5.5) proves (5.3). The ordinate formula (5.4) is the corresponding Landau–Hlawka–Ford–Zaharescu resonance formula; see [1]. Unique factorization gives the final assertion. □

Corollary 5.2 (Exact linear bridge)**.** *For every fixed shell R,*

$$\sum_{(a,b)\in\mathcal{S}_R} w_{a,b;R}^{(p,q)} e^{-i(a\vartheta_p + b\vartheta_q)} F_T\Big(u_{a,b}^{(p,q)}\Big) \tag{5.7}$$

$$= \frac{1}{N(T)} \sum_{0<\gamma\le T} \Big(K_{R,\vartheta}^{(p,q)}(\gamma - i\delta_\rho) - K_{R,\vartheta}^{(p,q)}(\gamma)\Big) \tag{5.8}$$

$$= o_R\Big(\frac{1}{\log T}\Big). \tag{5.9}$$

Proof. The equality of (5.7) and (5.8) follows by expanding the definitions. Proposition 5.1 makes the numerator in (5.8) equal to $o_R(T)$. The Riemann–von Mangoldt formula gives $N(T) \asymp T\log T$ [2]. □

6. A positive finite-height energy bridge

Let $m_T(\delta,\tau)$ be the multiplicity of the zero $1/2+\delta+i\tau$ for $0<\tau\le T$ and $\delta>0$. For $x>0$, define

$$D_x(\tau) = \sum_{\delta>0} m_T(\delta,\tau)\,(\cosh(\delta\log x) - 1)\,. \tag{6.1}$$

Functional-equation symmetry pairs δ with $-\delta$ at the same ordinate.

Let $\mathrm{d}m(\vartheta) = (2\pi)^{-2}\,\mathrm{d}\vartheta_p\,\mathrm{d}\vartheta_q$ be normalized Haar measure on $\mathbb{T}^2$. Define

$$\mathcal{E}_{R,T}^{\mathrm{full}} = \frac{1}{N(T)} \sum_{0<\gamma\le T} \int_{\mathbb{T}^2} \Big|K_{R,\vartheta}^{(p,q)}(\gamma - i\delta_\rho)\Big|^2\,\mathrm{d}m(\vartheta), \tag{6.2}$$

$$\mathcal{E}_{R,T}^{\mathrm{ord}} = \frac{1}{N(T)} \sum_{0<\gamma\le T} \int_{\mathbb{T}^2} \Big|K_{R,\vartheta}^{(p,q)}(\gamma)\Big|^2\,\mathrm{d}m(\vartheta), \tag{6.3}$$

$$\Delta_{R,T}^{(p,q)} = \mathcal{E}_{R,T}^{\mathrm{full}} - \mathcal{E}_{R,T}^{\mathrm{ord}}. \tag{6.4}$$

Theorem 6.1 (Exact energy bridge)**.** *For every finite shell,*

$$\mathcal{E}_{R,T}^{\mathrm{ord}} = Q_R^{(p,q)}, \tag{6.5}$$

$$\Delta_{R,T}^{(p,q)} = \frac{2}{N(T)} \sum_{0<\tau\le T} \sum_{\delta>0} m_T(\delta,\tau) \sum_{(a,b)\in\mathcal{S}_R} \Big(w_{a,b;R}^{(p,q)}\Big)^2 \Big(\cosh(2\delta u_{a,b}^{(p,q)}) - 1\Big) \tag{6.6}$$

$$= \frac{2}{N(T)} \sum_{(a,b)\in\mathcal{S}_R} \Big(w_{a,b;R}^{(p,q)}\Big)^2 \sum_{0<\tau\le T} D_{\exp(2u_{a,b}^{(p,q)})}(\tau). \tag{6.7}$$

If $\mathcal{S}_R$ contains an index $(a,b)\neq(0,0)$, then

$$\boxed{\Delta_{R,T}^{(p,q)} = 0 \quad\Longleftrightarrow\quad \operatorname{Re}\rho = \frac{1}{2} \text{ for every } 0<\operatorname{Im}\rho\le T.} \tag{6.8}$$

Proof. Orthogonality of the phase characters gives, for every real δ,

$$\int_{\mathbb{T}^2} \left|K_{R,\vartheta}^{(p,q)}(\gamma - i\delta)\right|^2 \, \mathrm{d}m(\vartheta) = \sum_{(a,b)\in\mathcal{S}_R} \left(w_{a,b;R}^{(p,q)}\right)^2 e^{2\delta u_{a,b}^{(p,q)}}. \tag{6.9}$$

Setting $\delta = 0$ proves (6.5). Pairing δ and $-\delta$ proves (6.6); Definition (6.1) gives (6.7). Every summand in (6.6) is nonnegative. A nonzero displacement makes every term with $u_{a,b}^{(p,q)} > 0$ strictly positive, proving (6.8). □

Define the finite-height curvature

$$F_T''(0) = \frac{2}{N(T)} \sum_{0<\tau\le T} \sum_{\delta>0} m_T(\delta,\tau)\delta^2. \tag{6.10}$$

Corollary 6.2 (Coercive curvature bound)**.** *Let*

$$M_{2,R}^{(p,q)} = \sum_{(a,b)\in\mathcal{S}_R} \left(w_{a,b;R}^{(p,q)}\right)^2 \left(u_{a,b}^{(p,q)}\right)^2. \tag{6.11}$$

Then

$$\boxed{\Delta_{R,T}^{(p,q)} \ge 2M_{2,R}^{(p,q)} F_T''(0).} \tag{6.12}$$

Proof. Apply $\cosh t - 1 \ge t^2/2$ termwise in (6.6). □

Corollary 6.3 (Explicit $(5,7)$ bridge)**.** *For the limit weights*

$$w_{a,b;\infty}^{(5,7)} = \frac{24}{35} 5^{-a} 7^{-b}, \tag{6.13}$$

set

$$J_{5,7}(\delta) = \sum_{a,b\ge 0} \left(w_{a,b;\infty}^{(5,7)}\right)^2 e^{2\delta(a\log 5 + b\log 7)} = \frac{(24/35)^2}{(1-5^{2\delta-2})(1-7^{2\delta-2})}. \tag{6.14}$$

For $|\delta| < 1/2$ this is finite, and $J_{5,7}(0) = 1/2$. The limiting energy difference satisfies

$$\Delta_{\infty,T}^{(5,7)} = \frac{1}{N(T)} \sum_{0<\gamma\le T} \left(J_{5,7}(\delta_\rho) - \frac{1}{2}\right) \tag{6.15}$$

and

$$\boxed{\Delta_{\infty,T}^{(5,7)} \ge 0.204533768513927\ldots \, F_T''(0).} \tag{6.16}$$

More exactly,

$$M_{2,\infty}^{(5,7)} = \frac{13}{576}(\log 5)^2 + \frac{25}{2304}(\log 7)^2 + \frac{1}{1152}\log 5 \log 7 = 0.102266884256963\ldots. \tag{6.17}$$

Proof. Equation (6.14) is the product of two geometric series. Passing to the limit shell in (6.9) gives (6.15). Direct differentiation of the geometric series, or the identities

$$\sum_{k\ge 0} r^k = \frac{1}{1-r}, \quad \sum_{k\ge 0} k r^k = \frac{r}{(1-r)^2}, \quad \sum_{k\ge 0} k^2 r^k = \frac{r(1+r)}{(1-r)^3}, \tag{6.18}$$

gives (6.17). Corollary 6.2 then gives (6.16). □

7. What the contraction determines

Theorem 7.1 (Finite horizontal perturbations preserve the asymptotic inputs)**.** *Let a symmetric zero multiset satisfy the Riemann–von Mangoldt count, Hlawka equidistribution, the Landau formula* (5.5)*, and the mean-square contraction* (4.10)*. Adjoin, at one fixed positive ordinate τ, the symmetric pair*

$$\frac{1}{2}+\delta+i\tau, \qquad \frac{1}{2}-\delta+i\tau, \qquad 0<\delta<\frac{1}{2}, \tag{7.1}$$

together with the conjugate pair at negative ordinate. Then all four asymptotic statements remain valid, while for every $T \geq \tau$,

$$F_T''(0) > 0, \qquad \Delta_{\infty,T}^{(5,7)} > 0. \tag{7.2}$$

Consequently, the mean-square contraction and the stated Mangoldt asymptotics do not imply the energy equality $\Delta_{\infty,T}^{(5,7)} = 0$.

Proof. The insertion changes $N(T)$ by at most two and changes every fixed normalized zero-phase average by $O(1/N(T))$. It therefore preserves the Riemann–von Mangoldt and Hlawka asymptotics and the limit in (4.10). For every fixed $x > 1$, its contribution to the Landau sum is

$$x^{1/2+\delta+i\tau} + x^{1/2-\delta+i\tau} = 2x^{1/2}e^{i\tau\log x}\cosh(\delta\log x), \tag{7.3}$$

which is $O_x(1)$ and is absorbed by the error $O_x(\log T)$ in (5.5). The strict inequalities in (7.2) follow from (6.10) and Theorem 6.1. □

Proposition 7.2 (Energy transfer preserves the horizontal excess)**.** *For arbitrary complex branch amplitudes z_p, z_q,*

$$\left|\frac{z_p+z_q}{\sqrt{2}}\right|^2 + \left|\frac{z_p-z_q}{\sqrt{2}}\right|^2 = |z_p|^2+|z_q|^2. \tag{7.4}$$

Thus contraction of the averaged branch transfers the missing energy to the complementary branch. Applied to the complete zero branches, the conserved total is $Q_R^{(p,q)} + \Delta_{R,T}^{(p,q)}$, not $Q_R^{(p,q)}$ when a horizontal displacement is present.

Proof. Equation (7.4) is the parallelogram identity. The second assertion follows from the definition (6.4). □

8. Conclusion

The prime pair $(5, 7)$ has the exact zero-ordinate mean-square contraction factor $1/2$, and the moments of every fixed iterate are given by the torus integral (4.7). The isolated pointwise resonances do not obstruct this averaged contraction. The Mangoldt response is supported on the two prime-power axes; the mixed branches 5^a7^b, $a, b \geq 1$, have zero Mangoldt main term.

The linear identity (5.8) connects the kernel directly to the existing finite-height exponential difference F_T. Phase orthogonality gives the stronger positive energy identity (6.6). For $(5, 7)$ it yields the explicit coercive estimate (6.16), and vanishing of the energy difference is equivalent to critical-line containment up to height T.

Theorem 7.1 determines the logical relation between these statements: the zero-ordinate contraction and the stated Landau–Mangoldt asymptotics are unchanged by a finite symmetric horizontal perturbation, whereas the positive energy bridge responds immediately. Energy conservation transfers the corresponding excess between complementary branches but does not remove it.

References

[1] K. Ford and A. Zaharescu, On the distribution of imaginary parts of zeros of the Riemann zeta function, *J. Reine Angew. Math.* **579** (2005), 145–158.

[2] E. C. Titchmarsh, *The Theory of the Riemann Zeta-function*, 2nd ed., revised by D. R. Heath-Brown, Oxford University Press, 1986.